Experiment-21

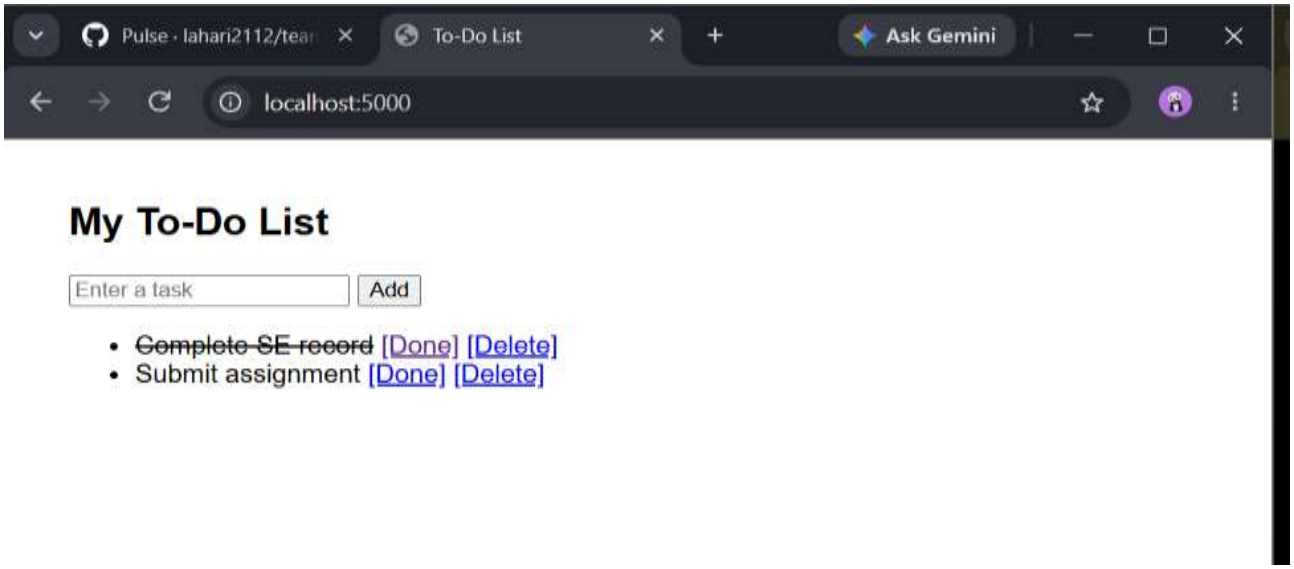
CONTAINERIZING A PYTHON FLASK TO-DO APPLICATION USING DOCKER

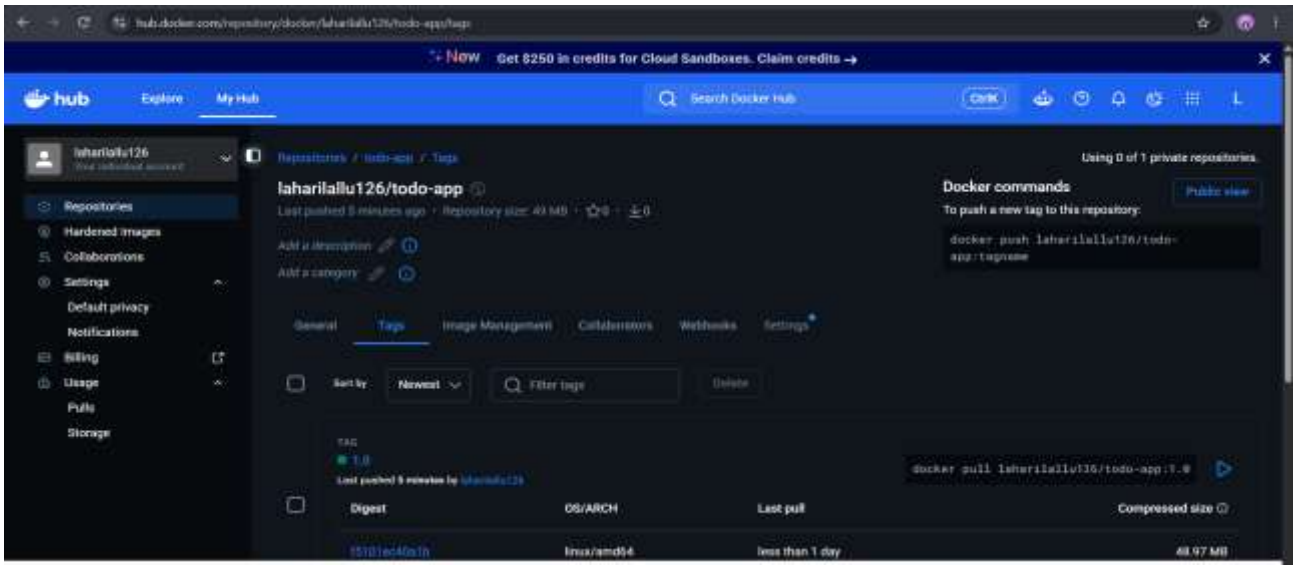
AIM

To write a simple Python Flask To-Do list application, containerize it with a Dockerfile, build and run it locally, test its functionality inside the container and push the image to Docker Hub.

Procedure

1. Created a Python Flask To-Do application to add, view, update and delete tasks.
2. Created the required project files, including app.py and requirements.txt.
3. Added Flask dependencies to the requirements file.
4. Created a Dockerfile to define the application environment and configuration.
5. Built the Docker image using the docker build command.
6. Ran the application inside a Docker container using the docker run command.
7. Mapped the container port to the host port to access the application.
8. Opened the application in a web browser using localhost.
9. Tested the To-Do application by adding and managing tasks.
10. Verified that the Flask application was running successfully inside the Docker container.





Result

Thus, the Python Flask To-Do application was successfully containerized using Docker and executed inside a Docker container. The application was accessed and tested through a web browser.